

# Wednesday Reading Assessment: Unit 3, Magnetic Forces and Fields

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## 1 Memory Bank

- $\hat{i} \times \hat{j} = \hat{k}$ ,  $\hat{j} \times \hat{k} = \hat{i}$ ,  $\hat{k} \times \hat{i} = \hat{j}$  ... The direction of the *cross-product* follows this pattern. Reversing the order of any two vectors introduces a minus sign.
- $\hat{i} \times \hat{i} = 0$ ,  $\hat{j} \times \hat{j} = 0$ ,  $\hat{k} \times \hat{k} = 0$  ... The *cross-product* is zero when both vectors are parallel.
- $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$  ... The magnitude of the cross-product of two vectors  $\vec{a}$  and  $\vec{b}$ , given the angle  $\theta$  between them.
- $\vec{F} = I\vec{L} \times \vec{B}$  ... The Lorentz force, by a magnetic field  $\vec{B}$  on a *current*  $i$  of length and direction  $\vec{L}$ .

## 2 Voltmeters and Ammeters

1. Suppose the ammeter in Fig. 1 has an internal resistance  $r = 10\Omega$ , and a “shunt” resistance of  $R = 0.1\Omega$ . The circuit has  $R_1 = 1\text{ k}\Omega$ ,  $R_2 = 1\text{ k}\Omega$ ,  $\mathcal{E} = 12\text{V}$  and  $r_b = 1\Omega$ . (a) Calculate the total resistance of the ammeter,  $R_A$ . (b) Calculate the total current flow from the battery,  $I_{\text{total}}$ . (c) What is the decrease in voltage across the ammeter? (d) What is the current flow to G, the galvanometer,  $I_G$ ?

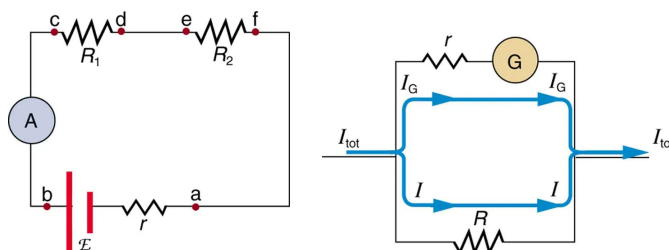


Figure 1: (Left) An ammeter in series with  $R_1$  and  $R_2$ , and a battery with emf  $\mathcal{E}$  and internal resistance  $r_b$ . (Right) The internal structure of the ammeter, with galvanometer G, internal resistance  $r$  and shunt resistance  $R$ .

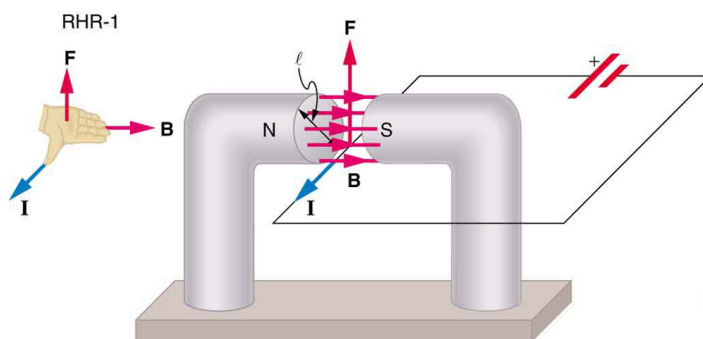


Figure 2: A permanent magnet creates a magnetic field to the *right*, while a voltage pushes a current *out of the page*. The force is measured to occur in the *upward* direction.

## 3 The Cross-Product

1. Let  $\vec{a} = 4\hat{i}$ , and  $\vec{b} = 4\hat{j}$ . (a) Calculate  $\vec{a} \times \vec{b}$ . (b) What are the magnitude and direction of the result?
2. What is the angle between the two vectors in the previous problem?

## 4 Magnetic Force on a Wire

1. Consider Fig. 2, in which a current passes through a *magnetic field* generated by a permanent magnet. (a) If the length of wire in the B-field is  $L = 10\text{ cm}$ , the magnetic field is  $B = 10^{-1}\text{ Tesla}$ , the voltage is  $V = 24\text{ Volts}$ , and the effective resistance in the wire is  $R = 2\Omega$ , what is the force upwards on the wire? (b) If the wire was attached to a scale, and the scale has a mass of  $m = 24\text{ grams}$  on it, what would the scale read if we turn on the current?<sup>1</sup>

<sup>1</sup>Hint: it won't say 24 grams.